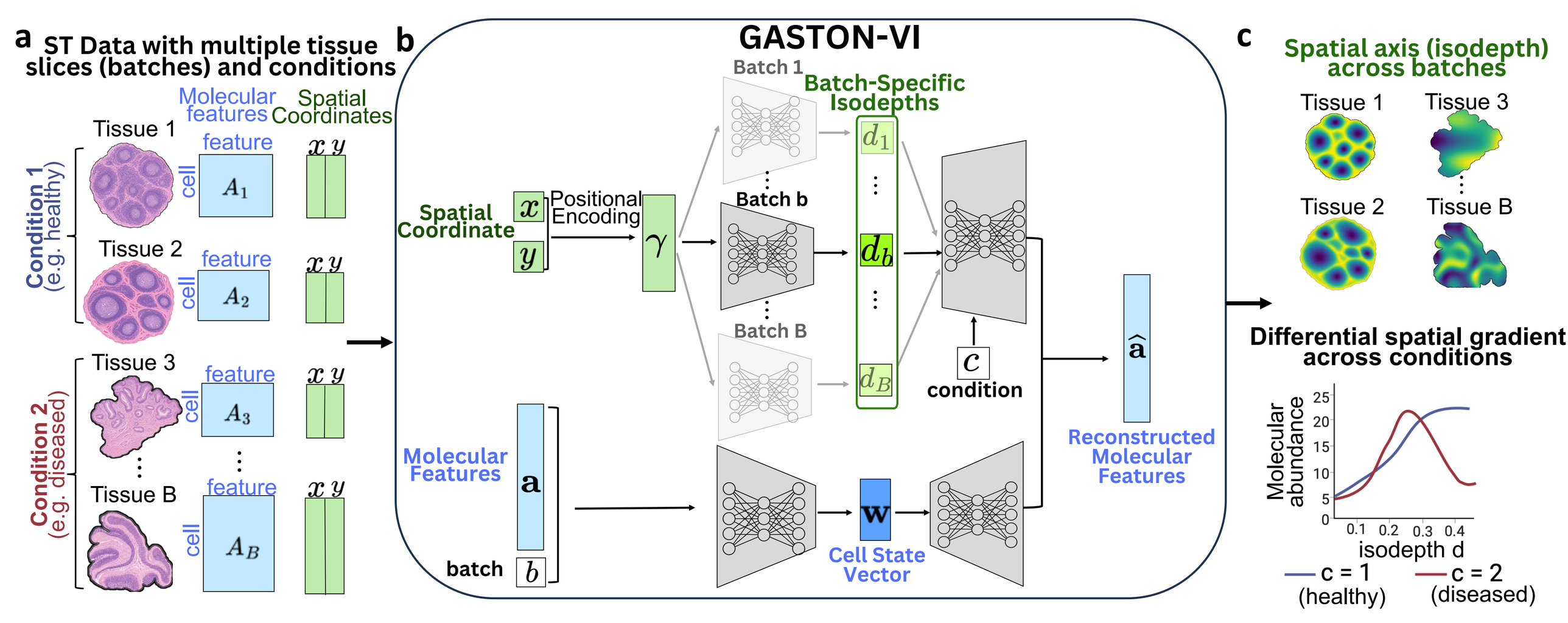

a
ST Data with multiple tissue slices (batches) and conditions
Molecular features
Spatial Coordinates
Condition 1 (e.g. healthy)
Tissue 1
Tissue 2
Condition 2 (e.g. diseased)
Tissue 3
Tissue B
feature
cell
A_1
A_2
A_3
A_B
x y
b
GASTON-VI
Batch 1
Batch-Specific Isodepths
Batch b
Batch B
Spatial Coordinate
Positional Encoding
γ
d_1
d_b
d_B
Molecular Features
$\mathbf{a}$
batch
b
$\mathbf{w}$
Cell State Vector
c
condition
$\hat{\mathbf{a}}$
Reconstructed Molecular Features
c
Spatial axis (isodepth) across batches
Tissue 1
Tissue 3
Tissue 2
Tissue B
Differential spatial gradient across conditions
Molecular abundance
25
20
15
10
5
0
0.1
0.2
0.3
0.4
isodepth d
c = 1
(healthy)
c = 2
(diseased)